

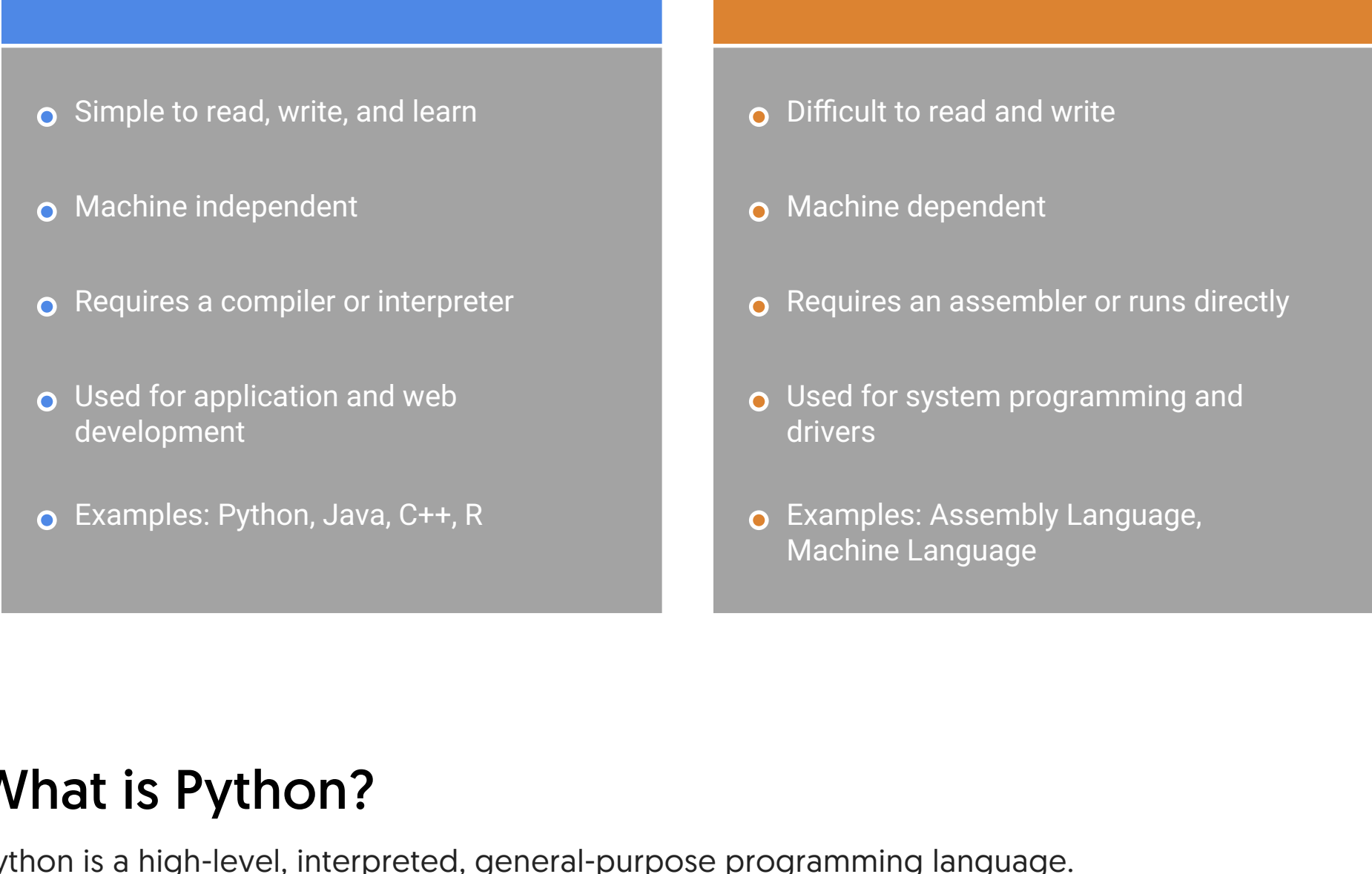
What is Programming?

Programming is the process of writing instructions that a computer can understand and execute.
A set of instructions written to perform a specific task is called a program.
Programming languages are used to communicate with computers.

Examples

- Calculate student marks
- Generate reports
- Build websites
- Predict sales using Machine Learning

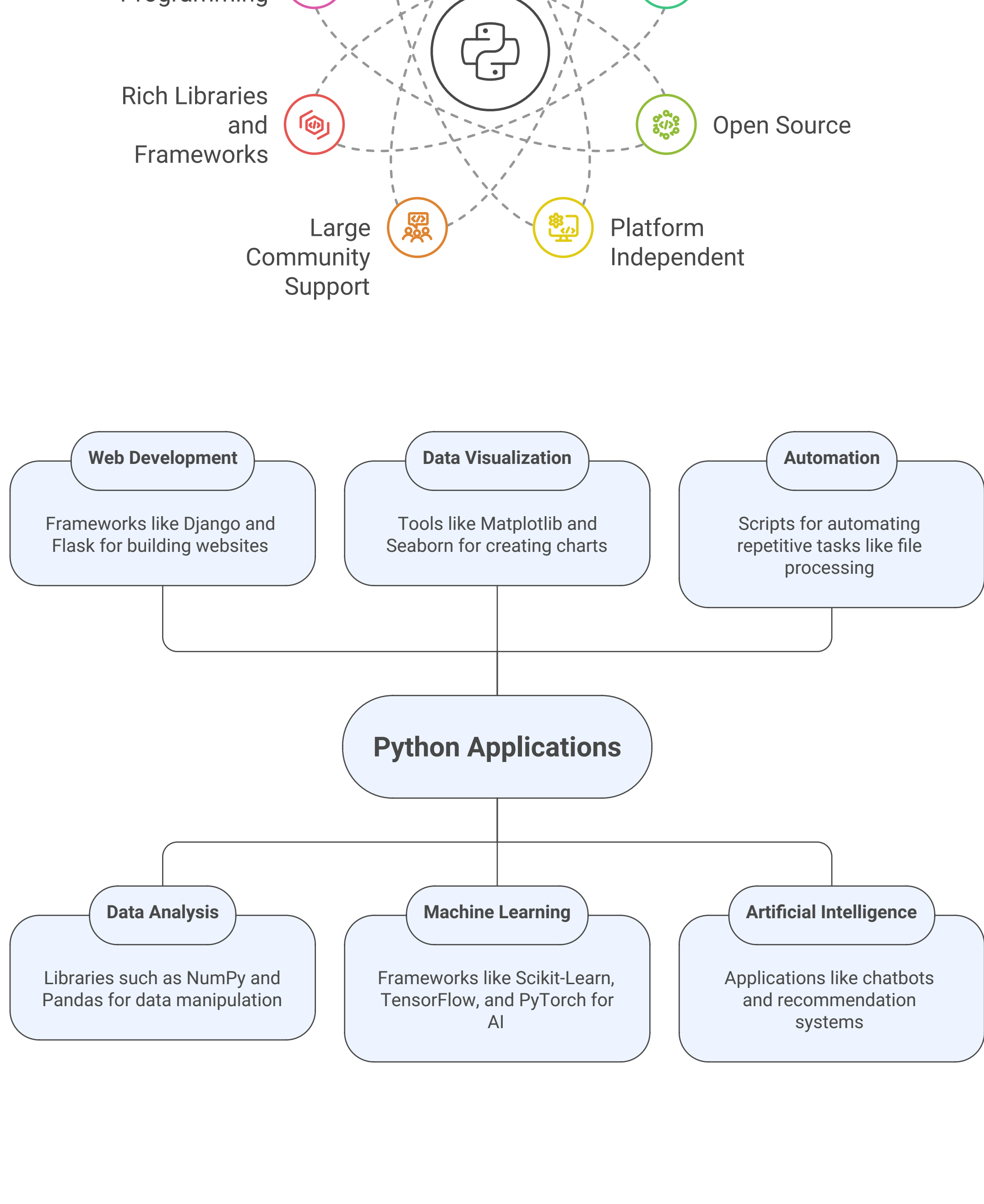
Types of Programming Languages



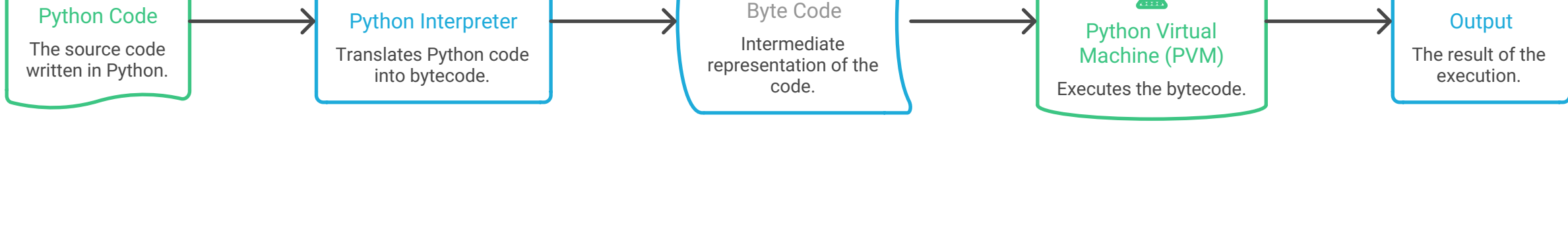
What is Python?

Python is a high-level, interpreted, general-purpose programming language.

Python's Key Features



Python Execution Flow



Setting Up Python Environment

Anaconda

Anaconda is a distribution that comes with:

- Python
- Jupyter Notebook
- JupyterLab
- Spyder
- Popular Data Science Libraries

Why Anaconda?

- Easy installation
- No need to install packages separately
- Widely used in Data Science

Jupyter Notebook

Jupyter Notebook is an interactive environment used to write and execute Python code.

Features

- Run code cell by cell
- Visualize outputs
- Supports Markdown
- Ideal for Data Analysis and Machine Learning

Google Colab

Google Colaboratory [Colab] is a cloud-based Jupyter Notebook environment.

Advantages

- No installation required
- Free GPU support
- Accessible from browser
- Easy sharing and collaboration

First Python Program

```
print("Hello World")
```

Output

```
Hello World
```

Comments

Comments are used to improve code readability.
Python ignores comments during execution.

Single Line Comment

```
# This is a comment
```

Multi Line Comment

```
"""
This is a
multi-line comment
"""
```

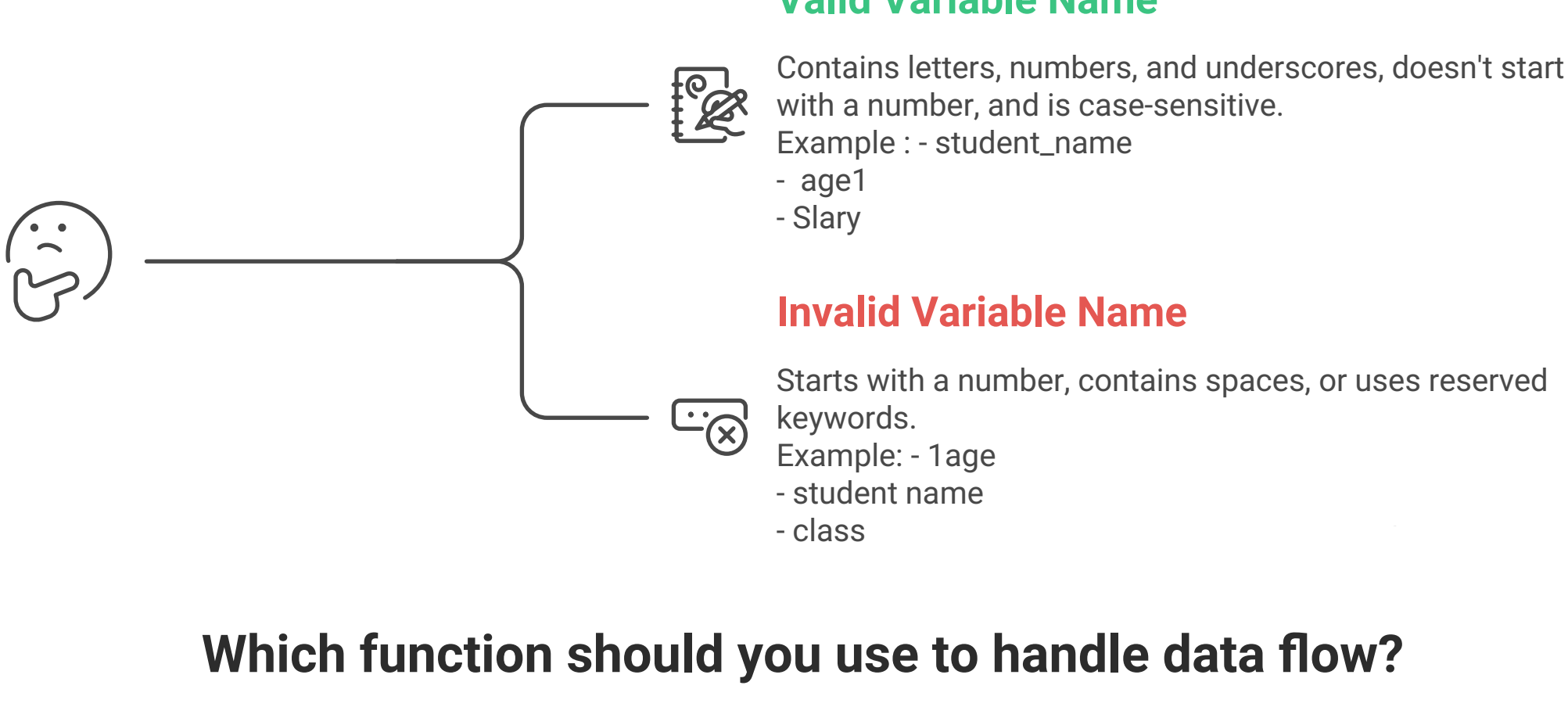
Variables

A variable is a name used to store data in memory.

Example

```
name = "John"
age = 25
salary = 50000
```

How to name a variable in Python?



Which function should you use to handle data flow?

Output Function

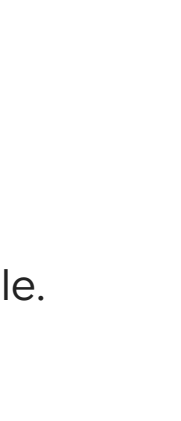


Displays information to the user

Uses the print command to show results

print("Hello")

Input Function



Captures data from the user

Uses the input command to receive values

name = input("Enter Name: ")

Data Types

Data type specifies the type of value stored in a variable.

Data Type	Description	Example
Integer (int)	Stores whole numbers.	'age = 25'
Float (float)	Stores decimal numbers.	'price = 99.99'
Complex (complex)	Stores real and imaginary values.	'x = 3 + 4j'
Boolean (bool)	Stores True or False.	'is_student = True'
String (str)	Stores textual data.	'name = "Python"'
NoneType	Represents absence of value.	'x = None'

Type Checking

Use type[] to identify data type.

```
x = 10
print(type(x))
```

Output

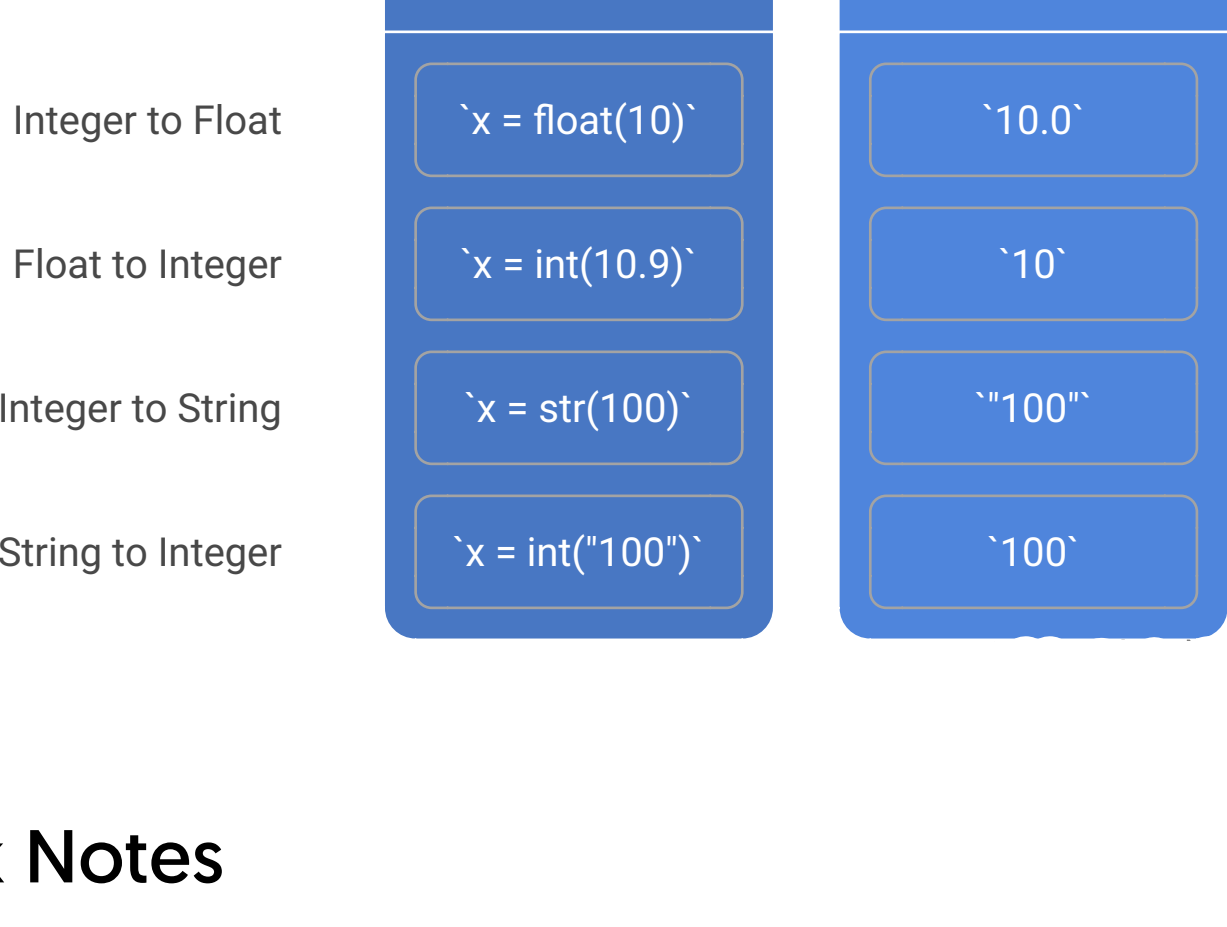
```
<class 'int'>
```

Example

```
print(type(10))
print(type(10.5))
print(type("Python"))
print(type(True))
```

Type Conversion

Type conversion is the process of converting one data type into another.



Quick Notes

Python is Case Sensitive

```
name = "John" # Both variables are different.
Name = "David"
```

Multiple Variable Assignment

```
a, b, c = 10, 20, 30
```

Same Value Assignment

```
x = y = z = 100
```

Swap Two Variables

```
a = 10
b = 20
a, b = b, a
```

Output

```
a = 20
b = 10
```

Practice Questions

1. What is the difference between a high-level language and a low-level language?
2. Why is Python called an interpreted language?
3. What is the purpose of Anaconda?
4. What is the difference between Jupyter Notebook and Google Colab?
5. Write a program to print your name.
6. Create variables to store:
 - Name
 - Age
 - Salary
7. Identify the data type:

```
10
10.5
True
"Python"
3+4j
None
```

8. Convert:

```
"100" → int
10 → str
20 → float
```

9. Write a program to take user input and display it.
10. What is the output?

```
x = 10
print(type(x))
```